

# BC004 · Flask & Safe Bots

Python Bootcamp · 100 Days — Days 49–64. Automation, Flask, Bootstrap, SQLAlchemy.

**For parents:** these are the key English coding words your child meets in this course, week by week. You don't need to code — just read a word together and ask your child to explain it back in their own words. The tiny example shows what it looks like in real code.

## W1

| WORD         | WHAT IT MEANS                    | TINY EXAMPLE                        |
|--------------|----------------------------------|-------------------------------------|
| automation   | code doing a manual task for you | <pre>auto-fill a form</pre>         |
| find_element | locate something on a web page   | <pre>driver.find_element(...)</pre> |

## W2

| WORD            | WHAT IT MEANS                  | TINY EXAMPLE                           |
|-----------------|--------------------------------|----------------------------------------|
| loop automation | repeating an action many times | <pre>for i in range(10): click()</pre> |
| wait            | pause until the page is ready  | <pre>time.sleep(2)</pre>               |

## W3

| WORD        | WHAT IT MEANS                  | TINY EXAMPLE                    |
|-------------|--------------------------------|---------------------------------|
| log to file | save a record of what happened | <pre>open("log.txt", "a")</pre> |

## W4

| WORD             | WHAT IT MEANS             | TINY EXAMPLE                 |
|------------------|---------------------------|------------------------------|
| dictionary tally | count things using a dict | <pre>counts[name] += 1</pre> |

## W5

| WORD         | WHAT IT MEANS                 | TINY EXAMPLE                |
|--------------|-------------------------------|-----------------------------|
| iterate rows | loop through each row of data | <pre>for row in rows:</pre> |

## W6

| WORD      | WHAT IT MEANS                   | TINY EXAMPLE                       |
|-----------|---------------------------------|------------------------------------|
| Flask     | a Python framework for websites | <pre>from flask import Flask</pre> |
| route     | a URL the app responds to       | <pre>@app.route("/")</pre>         |
| decorator | the @ line that adds behavior   | <pre>@app.route(...)</pre>         |
| app.run() | start the web server            | <pre>app.run(debug=True)</pre>     |

## W7

| WORD           | WHAT IT MEANS              | TINY EXAMPLE                              |
|----------------|----------------------------|-------------------------------------------|
| route variable | a changing part of the URL | <code>@app.route("/&lt;int:n&gt;")</code> |

## W8

| WORD        | WHAT IT MEANS                    | TINY EXAMPLE                                  |
|-------------|----------------------------------|-----------------------------------------------|
| return HTML | the text/page a route sends back | <code>return "&lt;h1&gt;Hi&lt;/h1&gt;"</code> |

## W9

| WORD            | WHAT IT MEANS                           | TINY EXAMPLE                               |
|-----------------|-----------------------------------------|--------------------------------------------|
| render_template | show an HTML template file              | <code>render_template("index.html")</code> |
| Jinja           | template language for variables in HTML | <code>{{ name }}</code>                    |

## W10

| WORD      | WHAT IT MEANS                  | TINY EXAMPLE                                   |
|-----------|--------------------------------|------------------------------------------------|
| Bootstrap | a ready-made CSS style library | <code>&lt;link href="bootstrap.css"&gt;</code> |

## W11

| WORD                 | WHAT IT MEANS                 | TINY EXAMPLE                           |
|----------------------|-------------------------------|----------------------------------------|
| component            | a reusable styled UI block    | <code>&lt;div class="card"&gt;</code>  |
| template inheritance | pages sharing one base layout | <code>{% extends 'base.html' %}</code> |

## W12

| WORD             | WHAT IT MEANS                     | TINY EXAMPLE                            |
|------------------|-----------------------------------|-----------------------------------------|
| form method POST | send form data to the server      | <code>&lt;form method="POST"&gt;</code> |
| request.form     | read submitted form data in Flask | <code>request.form['name']</code>       |

## W13

| WORD      | WHAT IT MEANS                        | TINY EXAMPLE                             |
|-----------|--------------------------------------|------------------------------------------|
| WTForms   | a library that builds & checks forms | <code>class LoginForm(FlaskForm):</code> |
| validator | a rule a form field must pass        | <code>DataRequired()</code>              |

## W14

| WORD     | WHAT IT MEANS             | TINY EXAMPLE                  |
|----------|---------------------------|-------------------------------|
| read CSV | load rows from a CSV file | <code>csv.reader(file)</code> |

**W15**

| WORD       | WHAT IT MEANS                      | TINY EXAMPLE                        |
|------------|------------------------------------|-------------------------------------|
| database   | organized storage for app data     | <code>movies table</code>           |
| SQLAlchemy | use a database with Python classes | <code>class Movie(db.Model):</code> |
| query      | ask the database for data          | <code>Movie.query.all()</code>      |

**W16**

| WORD     | WHAT IT MEANS                     | TINY EXAMPLE                              |
|----------|-----------------------------------|-------------------------------------------|
| CRUD     | Create, Read, Update, Delete data | <code>add / view / edit / delete</code>   |
| order_by | sort database results             | <code>query.order_by(Movie.rating)</code> |